

The Residence Time Distribution Analysis via Convolutional Neural Networks: A Promising Approach

Tuana Oyuncu

Conventionally, RTD analysis involves an iterative process between examining experimental data and predicting an appropriate model. It often leads to an unsuitable selection of a model to describe the system, thus resulting in a back-and-forth process. This report aims to eliminate the steps of conventional RTD methodology and its drawbacks by introducing convolutional neural networks (CNNs) into RTD analysis, taking advantage of the learning capabilities of neural networks. The model being developed mainly aims to substitute the compartment model by providing a solution without the necessity of oversimplification. The model is based on the experimental set-up, which is subdivided into four, and improved with the tuned hyperparameters. The results suggest that the integration of CNNs into RTD assessment holds considerable promise for improving the description of RTD functions within reactor models while opening the field for further investigation.

1. INTRODUCTION

Residence Time Distribution (RTD) is a fundamental metric in chemical reactor design, providing insight into mixing behavior and process performance. It has become one of the key analytical approaches for elucidating the complex dynamics of processes, where macroscale reactor behavior is inherently driven by fluid dynamics. The theory underlying this method is that the individual elements of the fluid spend varying lengths of time within a reactor or system referred to as the residence time of the atoms in the reactor [1,2]. The analysis of RTD profiles provides key knowledge to determine the transport of reactants through the reactor with the understanding of macromixing (i.e. back-mixing), which affects the quality of the product, conversion rates, and overall reactor performance. Thus, it is necessary to have an in-depth understanding of the RTD for the design applications [3]. Moreover, RTD analysis plays a critical role in periodic transition kinetic methods, where the reactor undergoes periodic transitions between different operating conditions, inducing variations in reactor behavior that affect reaction kinetics and product formation to investigate the dynamics of kinetic processes. RTD analysis allows researchers to characterize the distribution of residence times of fluid elements in the reactor during the periodic transitions and, by analyzing how the RTD profiles are affected in response to

changes in operating conditions [4]. Subsequently, the decoupling of flow effects from the intrinsic reaction kinetics is critical for the reliable determination of the true kinetic parameters and reaction mechanism of the underlying chemical processes.

Three main stages can be considered for the conventional methodology of RTD analysis: (i) experimentation, (ii) qualitative analysis, and (iii) mathematical modeling [3]. Pulse injection and step tracer experiments are used in the experimental phase, which is the measurement of the RTD. In a pulse experiment, a tracer with a small concentration, $C(t)$, is suddenly injected as a single shot into the feed stream entering the reactor in an infinitely short amount of time and the effluent concentration is then measured as a function of time. Alternatively, step experiments involve altering the operating conditions of the reactor, typically the flow rate or concentration of a reactant, and measuring the response of the system over time via a tracer. As a second step, qualitative analysis is carried out on the resulting information from the concentration and time data, and functions are utilized at this stage to evaluate the experimental data to determine the flow and mixing behavior. For this purpose, concentration versus

time graphs, as shown in Figure 1, can be used to analyze the general behavior of the system.

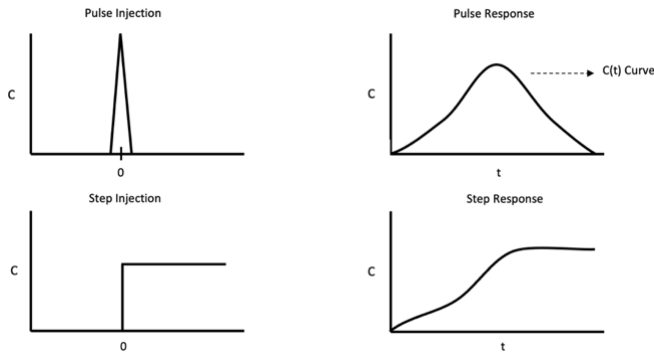


Figure 1. RTD measurements [2]

The RTD function describes the temporal dynamics of the different fluid elements in the reactor, denoted as $E(t)$. The function can be derived directly from the outlet tracer concentration for pulse experiments, while for step input it is defined by the cumulative distribution function, $F(t)$, obtained by integrating $E(t)$ over a series of time intervals [5,6]. The $E(t)$ can be derived directly from a pulse input, while the $F(t)$ is obtained from a step input due to the time interval characteristics of the experiments [7]. The description of $E(t)$ for pulse and step input is given by equations 1 and 2, respectively.

$$E_p(t) = \frac{C(t)}{\int_0^\infty C(t) dt} \quad (1)$$

$$E_s(t) = \frac{dF(t)}{dt} \quad (2)$$

Furthermore, $F(t)$ represents the successive contributions of the tracer elements that have been in the system for less than or equal to a specified period of time before leaving the system.

$$F(t) = \int_0^t E(t) dt \quad (3)$$

The qualitative phase is followed by the development of mathematical models designed to simulate and predict fluid flow and mixing dynamics. Both of these stages involve a cyclical process of iteratively identifying a model, implementing it, and assessing its accuracy through evaluation. It is likely that the system is initially associated with an inaccurate model and a scenario that often arises after the application and comparison of the behavior of the system. A variety of mathematical models are used, including

compartment methods, tank-in-series models, and dispersion models, as well as empirical approaches that are strongly based on experimental data. Compartment models divide the system into a series of interconnected compartments through which the substance flows. The model helps to provide a deeper insight into the flow dynamics by breaking down the system into manageable segments [1]. In addition, estimating the parameters of compartment models such as the number of compartments, their interconnections and the RTD function parameters for each compartment can be challenging, and fitting the model to experimental RTD data is often neither straightforward nor accurate. Particularly in complex systems, compartment models can be oversimplified and fail to capture all the intricacies such as dead zones, channeling, or back-mixing. The convolution integral technique is commonly used to allow better prediction of the system response to non-ideal step or pulse inputs for the corresponding models [8]. Convolution is a binary operation that combines two functions, representing the input stimulus and the output response of the system, to produce a third function. The resulting function is generated by integrating the pointwise multiplication of the original functions. In a manner of words, convolution "blends" one function with another to model the response of a system to an input signal in the signal processing with the output signal [9].

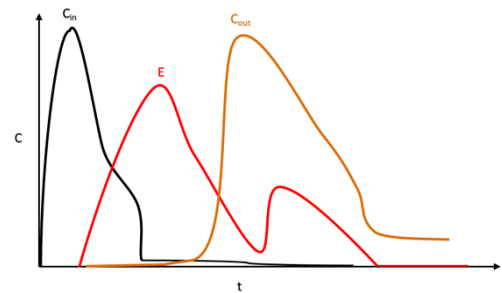


Figure 2. The representation of an input and output with a convolved signal [9]

The mathematics of the concept of convolution is rigorously defined as the integration over the entire spatial domain of a function, $f(x)$, multiplied by another function $g(x)$, evaluated at the displacement, $u - x$. This integral spans the variable, x , and overall dimensions, usually from negative infinity to positive infinity, which is particularly applicable in multidimensional cases [9]. Thus, it leads to a new function that depends on a new variable, u , exemplified by the formalized equations that explain the convolution process.

$$C(u) = f(x) \otimes g(x) = \int_{space} f(x) g(u - x) dx \quad (3)$$

$$= g(x) \otimes f(x) = \int_{space} g(x) f(u-x) dx \quad (4)$$

However, in chemical reactor design, where time is the independent variable, the lower limit of the integration is often set to zero and the upper limit of the integration is defined by a specific time, t , when computing the convolution. Additionally, the mean residence time, τ , which indicates the average period of time that a particle spends inside the reactor, is used in the description of the convolution process, equation 5.

$$C(t) = \int_0^t f(\tau) g(t-\tau) d\tau \quad (5)$$

The approach can be extended to a series of compartments by recursively evaluating the entire RTD, and the investigation of a specific unit can be performed in a multi-unit system [1,11].

$$C_{out,1} = E_1 \otimes C_{in} \quad (6)$$

$$C_{out,2} = E_2 \otimes C_{out,1} = E_1 \otimes E_2 \otimes C_{in} \quad (7)$$

$$C_{out,3} = E_3 \otimes C_{out,2} = E_1 \otimes E_2 \otimes C_{in} \quad (8)$$

$$\begin{aligned} C_{out,i+n} &= E_{i+n} \otimes C_{out,i+n-1} \\ &= E_i \otimes \dots \otimes E_{i+n} \otimes C_{in} \end{aligned} \quad (9)$$

While the convolutional integral technique is considered a beneficial technique for predicting system responses in analysis, inherent variations within systems introduce a level of error into the results and the struggle to choose an appropriate model is still a concern. Therefore, Convolutional Neural Network (CNN) techniques are considered to be a promising solution to overcome the shortcomings of the traditional RTD methodology. CNNs are capable of performing impressively in computer vision tasks of all kinds, including object identification, image recognition, and image retrieval [12]. They harness the power of deep learning by applying multiple convolutional layers that autonomously learn hierarchical representations from input data. In general, the convolutional layers within CNNs convolve the input data, extracting spatial and temporal patterns that are essential for understanding system behavior over time. Within this regard, the kernel which determines the range of features that a CNN can learn, convolves with the input data at each spatial location. Subsequently, it performs computations between the kernel elements and the corresponding input elements, and the results are then aggregated to produce the output. Therefore, several hyperparameters are used in machine learning

techniques for adjusting the profile fitting. They include the learning rate, which controls the magnitude of parameter updates during optimization, and the epoch number, which represents the cumulative number of iterations over the entire data set during training [13-15]. Since CNNs have a remarkable ability to extract and learn important features automatically from raw input data, and their end-to-end learning capability enables direct learning from raw input to output predictions without requiring intermediate steps, this study proposes a cutting-edge approach to RTD analysis by combining it with CNNs for advancing, improving and eliminating the iterative cycle in the methodology. A comprehensive study of profile fitting within the experimental setup over different flow rates with different tracer concentrations is performed. In addition, the experimental setup has been partitioned into different positions to ensure a thorough capture of the RTD dynamics as well as the individual compartments. The study also systematically investigates the effect of hyperparameters on model performance in order to optimize the fitting accuracy of the RTD functions.

2. EXPERIMENTAL SETUP AND METHOD

2.1 Experimental Setup

The comprehensive experimental framework described by Meyer et al. is used for the details of the unit operations of the system [4]. In order to perform the RTD analysis, a step tracer experiment is integrated into the system. A gas mixture consisting of hydrogen, helium, and argon is introduced into the experiment. Hydrogen is used as a carrier, while argon is the preferred tracer due to its ability to mimic the behavior of the substances in the system. Furthermore, helium is used as a reference ion in the mass spectrometry (MS) analysis. Figure 3 illustrates the experimental setup which is segmented into different sections that are essential for profile characterization within each compartment to define the RTD across the entire setup and individually, especially for the reactor. Thus, the entire system is subdivided into four discrete compartments by positioning the capillary of the MS at varying locations within the setup, and experiments are performed individually with the position of the capillary altered. Correspondingly, (1) the RTD of the MS capillary as a first path; (2) the RTD of the first pipe in the system; (3) the profile of the point after the tubular reactor, and (4) the encapsulation of the entire system. As shown in Table 1, the introduction of an additional compartment into the measurement extends the time interval of the data recorded during the experiment.

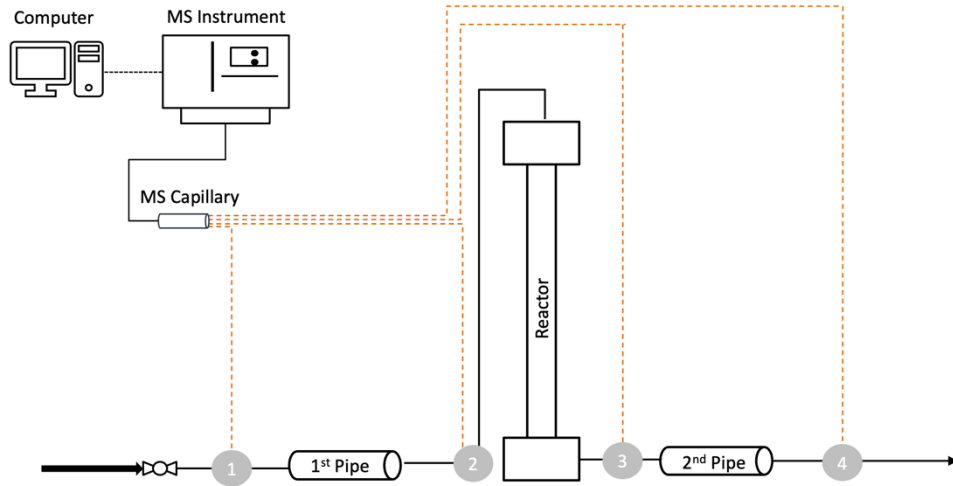


Figure 3. The schematic description of the experimental set-up

Table 1. Experiment duration according to the defined positions

Position	Time (s)
1	41
2	46
3	51
4	56

Figure 4 illustrates that the same step input is applied to the system to achieve similar step responses and the same maximum amount of tracer is reached over time but in different time domains for different experimental positions.

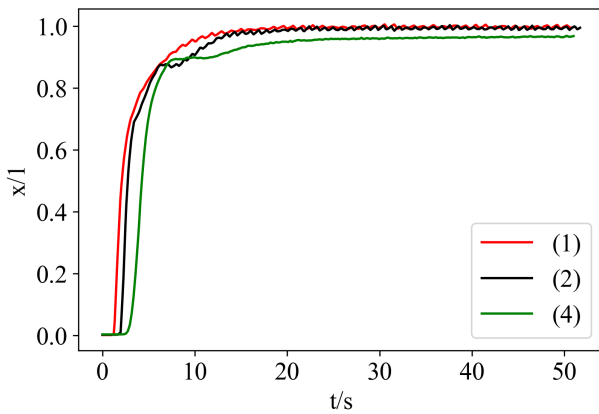


Figure 4. Volume fraction of the tracer over time at different positions, (1), (2), and (4)

2.2 Method and Data Evaluation

The model is developed on the basis of CNNs, and it is improved for RTD analysis. Within the CNNs structure, the model consists of a one-dimensional convolutional layer

defined in the PyTorch library. The convolutional layer convolves the input data with a kernel and computes the element-wise product between the kernel and the overlapping input elements. In this particular implementation, the input data consists of conventional step inputs, while the output function corresponds to experimental data obtained from RTD step experiments. The kernel, which plays a crucial role in feature extraction and pattern recognition, is a parameter that can be customized for analysis. Thus, the ability of the model to capture and learn complex relationships within the input-output data is influenced. Therefore, the replicate padding mode is chosen to maintain the integrity of the input signal by padding its edges with the nearest values [12]. Additionally, the convolutional layers are set to perform full convolution, aligning the padding size with the kernel size. The kernel size, which describes the scale of the kernel, specifies the degree of the region of the input with which the kernel interacts, thereby influencing the level of detail and information obtained by the convolution. Hence, the convolution operation encompasses the entire spatial domain of the input signal and prevents information loss during processing. Subsequently, the length of the output should be equal to the sum of the length of the input signal and the kernel size, plus one [8,12,15]. The discretization of the outputs is kept constant in the model, regardless of the position of the analysis being performed.

$$n_o = n_i + n_E + 1 \quad (10)$$

Since it is critical to ensure that each kernel element convolves with corresponding input elements, computes their product, and sums for output, the arithmetic of convolution in time is defined as Equation 11.

$$\Delta t_o = \Delta t_E + \Delta t_i \quad (11)$$

Since the length of the time vectors is not identical at each position, due to the variation in the development times of the RTD profiles with additional compartments, the following equality is combined with equations 10 and 11 to calculate the parameters of the kernel and the input signal, while ensuring proper temporal alignment between the signals.

$$\frac{n_E}{\Delta t_E} = \frac{n_i}{\Delta t_i} \quad (12)$$

$$n_i = (n_o - 1) \left(\frac{1}{1 + \frac{\Delta t_E}{\Delta t_i}} \right) \quad (13)$$

The calculations are performed individually for each position and the equations are kept the same for all experiments. However, due to the invariance of the parameters, calculations are not performed for different flow rates within the same point. It should be pointed out that the time specified for the convolution, Δt_E , and the number of discretizations for the output data, n_o , are arbitrarily chosen. Table 2 indicates the parameters specifically for position (1) and the rest of the results are illustrated in the supporting material, Table S1, Table S2, and Table S3.

Table 2. The model parameters for the position (1)

n_o	500
n_i	377
n_E	122
Δt_o	41 s
Δt_i	31 s
Δt_E	10 s

Moreover, a loss function is defined to specify the difference between the prediction of the model and the true output, while a test function evaluates the performance of the model on a data set to assess the ability of the model. The same data set is used for both training and testing the model, as well as for calculating the loss values, referred to as train/loss and test/loss. The underlying reason for the approach is to emphasize the availability of training data over the necessity of a separate, unseen dataset. Thus, the data set is not split, resulting in identical train/loss and test/loss values for this model.

In order to ensure the repeatability of the model results, the deterministic mode is implemented in the model to ensure the reproducibility of the model training and evaluation processes. The reproducibility is essential to ensure that the behavior of

the model is consistent throughout different runs and that the outcomes within the model can be accurately compared. Furthermore, the factor and patience parameters are introduced into the learning rate of the model to ensure the stability of the training process. The factor defines the degree to which the learning rate is decreased if the performance of the model becomes stagnant, while the patience parameter defines the number of epochs to have waited until the learning rate is decreased if there is no performance improvement. In this way, the model is provided with adequate time to become tuned to the current learning rate before any adjustments are implemented, thereby preventing early reductions that could hinder training progress.

3. RESULTS AND DISCUSSION

In order to improve the predictive accuracy of the model, the hyperparameters which are epochs number, and the learning rate, are examined before the profiles are evaluated, thus, allowing the model to better capture the underlying trends in the data.

3.1 Variation of Learning Rate

The investigation of the learning rate over the range from 10 to 10^{-6} is used to understand the response of the model to hyperparameter variation. Generally, lower learning rate values are beneficial in avoiding oscillations around the optimal solution, while allowing a cautious approach to convergence [12-14]. Conversely, higher learning rates accelerate convergence but run a significant risk of overshooting the loss function minimum which can cause instability during the training and affect overall performance. Therefore, managing the balance between convergence speed and stability becomes imperative in determining the most effective learning rate. For this model, it is observed that a higher learning rate means a greater negative effect, because of the resulting higher test/loss value. Therefore, according to Figure 6, the optimum approach is to select a relatively lower learning rate, mainly because it tends to be more stable with better fittings although the slight decrease in tes/loss is seen between 10 and 10^{-3} . However, it should be pointed out that at lower learning rates, oscillatory behavior remains during training, with slight variations in model accuracy. The corresponding observation can be found in the Supporting material, Figure S4. Since the precision of the model remains the same between 10^{-3} and 10^{-4} , with a greater oscillation in the 10^{-4} learning rate, the learning rate of the model is selected to be 10^{-3} to indicate the complex interplay between learning rate dynamics and model performance.

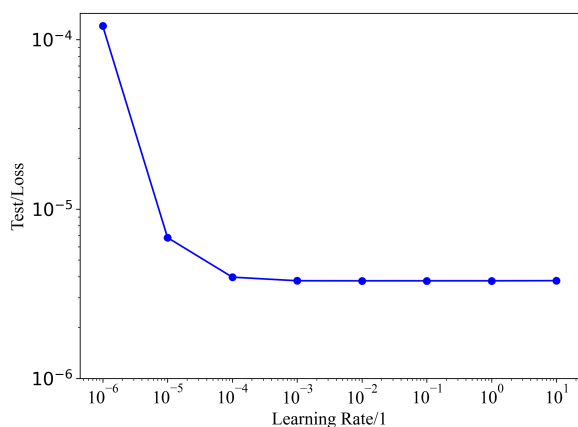


Figure 6. The responses of the model according to varied learning rate

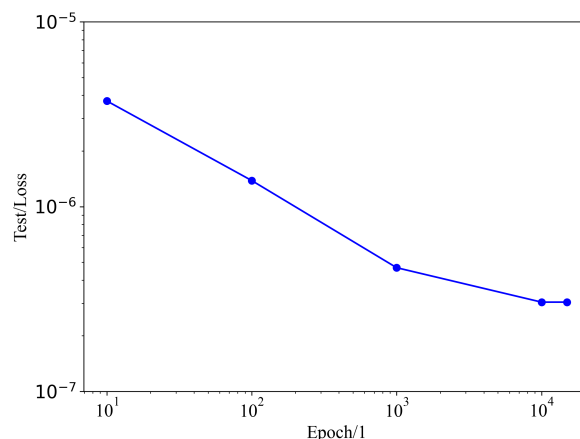


Figure 5. The responses of the model according to different epoch numbers

3.2 Variation of Epoch Number

The influence of the number of epochs, a hyperparameter that indicates the total number of cycles in which the entire data set is iteratively passed through the neural network during training, is investigated [12-14]. The number of epochs ranges from 10 to 15,000, with individual runs being carried out at different intervals, as shown in Figure 5. The model undergoes updates to its internal parameters at each epoch, resulting in an improvement in the fit, so that higher numbers of epochs are initially correlated with lower test/loss values, indicating an improvement in the accuracy of the model. However, there must be a balance between computational resources and model performance since the model might need more iterations, and therefore more computational resources are in demand in terms of processing power and memory. Additionally, as the number of epochs increases, the model is likely to converge to a local minimum in the loss landscape and further iterations may not significantly alter the model. According to Figure 5, the local minimum has been reached at 10,000 epochs and the further increment does not result in a lower test/loss value, conversely leading to the same model accuracy. Therefore, identifying 10,000 as the optimal number of epochs provides a compromise between achieving low training/loss, and computational efficiency, while ensuring robustness to model performance without excessive computational overhead.

3.3 Result of the System

The general behavior of the model and the effect of additional compartments in the system are analyzed at constant concentrations with varied spatial positioning. Even though the primary objective of the study was to investigate the four pre-defined points within the system for different concentration conditions, several failures occurred while processing the data from the third position and the highest concentration at the last point, rendering it unsuitable for inclusion in the study. However, the underlying concept remains valid and applicable to the research investigation.

All experiments are performed in accordance with the identified best parameters of learning rate and number of epochs.

3.3.1 Results and discussion of the position (1)

The investigation of the model with varying concentrations at each point begins with the RTD analysis of the capillary. It represents the behavior of a single compartment.

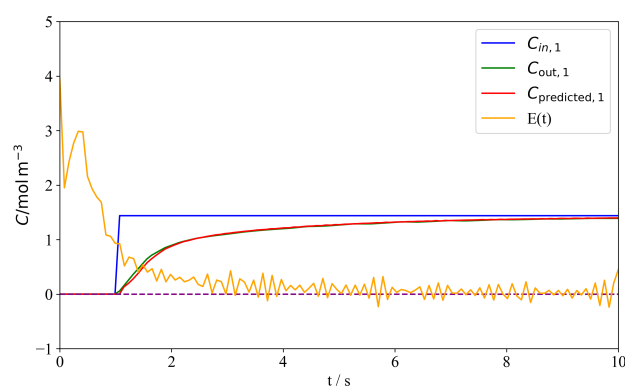


Figure 7. The illustration of the convolution function with the input and output signals at the first point.

According to Figure 7, agreement between the predicted concentration profile, corresponding to the cumulative residence time distribution function, and the experimental concentration profile is observed. Since the concentration profiles, both model output and experimental data, converge to the steady state of the step input, this indicates the ability of the model to accurately capture the dynamic and steady-state behavior of the system while ensuring close agreement.

It is assumed that the tracer is introduced into the system at $t=1$ s, with an ideal step. The input function is carefully tuned to match the experimentally determined tracer concentration. The sudden rise in the output function at precisely $t=1$ s indicates the approximately instantaneous response within the setup, describing the RTD of the MS capillary, which is characterized by the absence of dead volume. From Figure 7, the curve of $C_{\text{predicted},1}$ closely follows the trend of $C_{\text{out},1}$, indicating the ability of the model to accurately characterize the system response without the need for preliminary mathematical model investigation and subsequent iteration.

Since the first data point has a relatively short residence time compared to the other locations where that experiment is conducted, it results in a pronounced decay in the $E(t)$ function. The presence of it indicates a significant fraction of the tracer has experienced relatively short residence times in the intended volume. However, over time, $E(t)$ must eventually decrease to zero, obeying the essential probability normalization condition that the total area under the curve represents the cumulative probability and equals 1. According to Figure 7, although the $E(t)$ function decreases and oscillates around zero, it can be observed from the baseline, $x=0$, that the function oscillates around it with mirrored signals. Hence, despite the presence of noise in the system, it can be assumed that the function goes and remains zero.

3.3.2 Results and discussion of the position (2)

Figure 8 illustrates the result of the model for the more complex system response, which includes not only the RTD analysis of one compartment but also the time that the tracer spends in the MS capillary and the tube before the reactor. However, the model is still effective in capturing the system response, since $C_{\text{predicted},2}$ follows the same trend as $C_{\text{out},2}$ almost without deviation, although the system is becoming more complicated with two compartments.

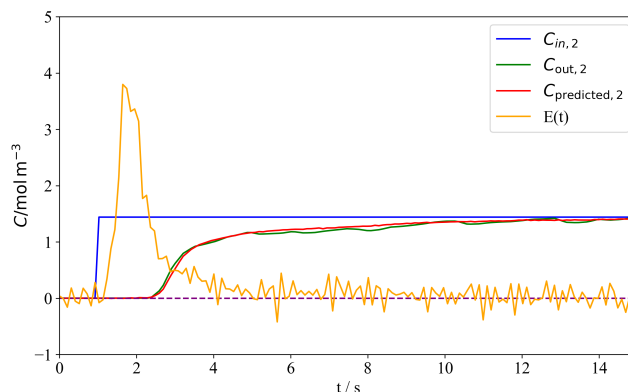


Figure 8. The representation of the convolution function with the input and output signals at the first point.

In comparison to the RTD analysis of position (1), a delay time is present in the system with the delayed increase in concentration profiles, despite the tracer being introduced into the system in the same method as initially. According to Figure 8, the concentration data which is taken during the experiment, $C_{\text{out},2}$, starts to increase by approximately $t \approx 2$ s. The reason for this is that the additional compartment leads to the tracer spending a longer period of time in the system, thereby causing the observed delay. Therefore, different kernel periods are specified for individual profiles to account for possible lag times caused by the addition of a component.

The cumulative residence time distribution function of both the experimental and predicted profiles shows a synchronous onset of increment and converges to a steady state over time. Furthermore, the $E(t)$ function is still predictable for this condition and in line with theory, reaching zero after decay.

3.3.2 Results and discussion of the position (4)

The representation of the RTD analysis of the whole system is described by point (4), which is located at the end of the set-up where the tracer passes through the multiple compartments including the reactor.

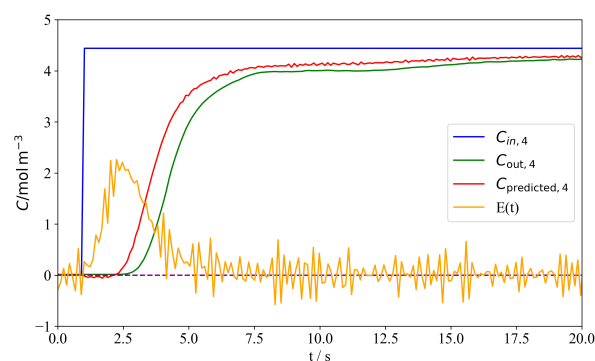


Figure 9. The illustration of the convolution function with the input and output signals at the first point.

The underfitting profile is observed for longer residence time profiles regardless of the tuned hyperparameters, as shown in Figure 9. Since the presence of additional compartments within the system introduces complexity to both data acquisition and processing, it potentially enhances the presence of noise and inconsistency in the data set. Due to the increased complexity, the neural network faces challenges as it struggles to capture patterns with the noise. Thus, the neural network can fail to fully capture the intricate relationships of the data, leading to suboptimal fitting of the output function. Even though there are deviations in the model output prediction, the model is still able to predict the $E(t)$ function with a decay reaching a zero value. However, this is associated with a higher oscillation because of the oscillations in the predicted profile.

Furthermore, the observed delay time, the time taken for the concentration to reach each measurement point, is consistent with the increased element in the experiment site since the longest response is observed at the last measurement point according to Figure 9. Therefore, a longer kernel time which is 20 s is defined for the model of the point (4).

3.4 Variation of Tracer Concentration and Flow Rate

Since it is critical to ensure that the tracer itself does not significantly affect or alter the flow behavior and properties of the bulk material, the variation in tracer concentration is assessed by varying flow rates with increasing amounts of hydrogen in the system. Figure 10 represents the total flow rates of 100, 150, and 200 mLmin⁻¹; the composition of the flows can be found in the supporting material, Table S4, Table S5, and Table S6. According to Figure 10, the deviations are present for each profile and the largest variation is seen in the highest flow response which is defined in Figure 10 (c). The convolved results indicate that a decreased concentration of the tracer, corresponding to higher flow rates, tends to generate increased noise levels within the recorded data set, resulting in the underfitting of profiles associated with higher flow rates, Figure S5. Conversely, an analogous behavior can be observed for low flow rate cases, where higher tracer concentrations can also lead to challenges in model fitting. Hence, a critical relationship exists to determine optimum operating parameters between the conditions of tracer concentration and flow rate to ensure robust model fitting and predictive accuracy within the system.

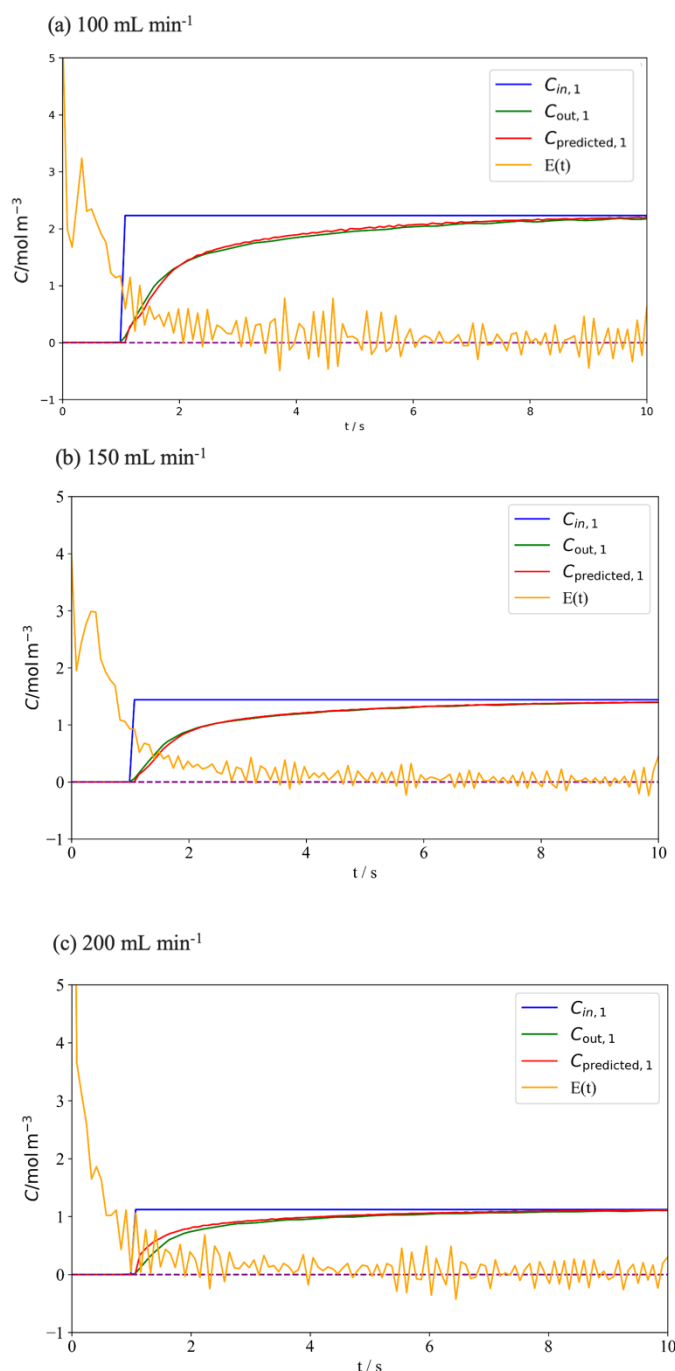


Figure 10. The model response for the varied flow rate, 100, 150, and 200 mLmin⁻¹

4. CONCLUSION and FUTURE WORK

This report demonstrates the integration of CNNs into RTD analysis, aiming to replace the conventional RTD methodology by eliminating the iterative process between qualitative and quantitative analysis steps. The model particularly targets the compartmental method by eliminating unrealistic assumptions and oversimplifications. The developed model is based on an

experimental setup which is segmented into four compartments and investigated at different flow rates. Furthermore, the hyperparameters affecting the accuracy of the model are investigated and it is found that the model provides a consistent output with C_{in} at 10,000 epochs and a learning rate of 10^{-3} . The developed model successfully captures the system response for the one-compartment and two-compartment systems at positions (1) and (2), respectively. However, it does not adequately predict the RTD of the overall multi-compartment system, due primarily to the limitations of the model and inconsistencies in the experimental data. Nevertheless, this innovative approach has the potential to advance chemical reaction engineering methods and improve reactor design applications with further research and improvement of CNN-based RTD analysis. Therefore, it is proposed that the model should be further developed with an increased amount of data for each specific compartment and the inclusion of point (3), which represents the RTD of the experimental setup after the reactor. Subsequently, the deconvolution technique will be investigated on the model to study the $E(t)$ of the reactor itself, and thus to enable it to be used for reactor simulations. During the investigations, conventional compartment models and their accuracy will be examined for a valid comparison between the conventional method and the proposed CNN-based method. In addition, future development will focus on improving the model through hyper-parameter optimization and investigating the effects on the improved model.

Supporting Information

The corresponding profiles, which have been examined and are mentioned in the report, are provided in the supporting material.

Acknowledgments

The author of this report would like to thank Prof. Dr.-Ing. Robert Güttel, M.Sc. Hannes Stagge and M.Sc. Max Gäßler for their guidance and support.

Nomenclature

Abbreviation

Abbreviation	Description
RTD	Residence Time Distribution
CNN	Convolutional Neural Network
MS	Mass Spectrometry

Subscripts

Subscripts	Description
o	Output
i	Input
E	Kernel
p	Pulse
s	Step

Symbols

Symbols	Unit	Descriptions
C	Mol m ⁻³	Concentration
E	1	RTD function
F	1	Cumulative distribution function
K	1	Kernel
n	1	Length of a vector
t	s	Time
x	1	Volume fraction
Δ	1	Delta
τ	s	Mean residence time

REFERENCES

1. Levenspiel, O. (1998). *Chemical reaction engineering*. John Wiley & sons.
2. Fogler, H. S. (2020). *Elements of chemical reaction engineering*. Pearson.
3. Nechita, M. T., Suditu, G. D., Puișel, A. C., & Drăgoi, E. N. (2023). Residence Time Distribution: Literature Survey, Functions, Mathematical Modeling, and Case Study—Diagnosis for a Photochemical Reactor. *Processes*, 11(12), 3420.
4. Meyer, D., Friedland, J., Schumacher, J., & Güttel, R. (2021). The periodic transient kinetics method for investigation of kinetic process dynamics under realistic conditions: Methanation as an example. *Chemical Engineering Research and Design*, 173, 253-266.
5. Naor, P., & Shinnar, R. (1963). Representation and evaluation of residence time distributions. *Industrial & Engineering Chemistry Fundamentals*, 2(4), 278-286.
6. Safaei, H., Falamaki, C., & Sohrabi, M. (2023). Applying a new approach to predict the residence time distribution in impinging streams reactors. *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects*, 45(3), 9569-9576.
7. Kanse Nitin, G., & Dawande, S. D. (2012). RTD Studies in Plug Flow Reactor and its Simulation with Comparing NonIdeal Reactors. *Research Journal of Recent Sciences*
8. Conesa, J. A. (2020). Convolution and Deconvolution of Residence Time Distribution Curves in Reactors. *Chemical Reactor Design: Mathematical Modeling and Applications*, 1.
9. Rodrigues, A. E. (2021). Residence time distribution (RTD) revisited. *Chemical engineering science*, 230, 116188.
10. Meyer, D., Friedland, J., Schumacher, J., Gäßler, M. G., & Güttel, R. (2022). Hydrogenation of CO/CO₂ mixtures under unsteady-state conditions: Effect of the carbon oxides on the dynamic methanation process. *Chemical Engineering Science*, 250, 117405.
11. Engisch, W., & Muzzio, F. (2016). Using residence time distributions (RTDs) to address the traceability of raw materials in continuous pharmaceutical manufacturing. *Journal of pharmaceutical innovation*, 11, 64-81.
12. Dumoulin, V., & Visin, F. (2016). A guide to convolution arithmetic for deep learning. *arXiv preprint arXiv:1603.07285*.
13. Zeiler, M. D., & Fergus, R. (2014). Visualizing and understanding convolutional networks. In *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part I* 13 (pp. 818-833). Springer International Publishing.
14. Yosinski, J., Clune, J., Nguyen, A., Fuchs, T., & Lipson, H. (2015). Understanding neural networks through deep visualization. *arXiv preprint arXiv:1506.06579*.
15. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Convolutional networks. *Deep learning*, 2016, 330-372.
16. Dantas, J. A., Pegoraro, P. R., & Gut, J. A. (2014). Determination of the effective radial mass diffusivity in tubular reactors under non-Newtonian laminar flow using residence time distribution data. *International Journal of Heat and Mass Transfer*, 71, 18-25.